Part 1

Investment Products

CHAPTER 10: Derivatives



Chapter Highlights

- When investing in the common shares of a company the value of that holding is generally based on the number of shares owned multiplied by the current market price.
- As the market price rises or falls, the value of the investor's holdings also rises or falls.
- Derivatives are a little different from holding shares or individual bonds.
- The value of a derivative is tied to something 'else' – an underlying asset such as a financial asset or commodity.
- It is the performance of this underlying asset that determines value.

Derivatives

- A derivative is a financial contract that has a value derived from, or dependent upon, the value of some other asset.
- The underlying asset of a derivative can be a financial asset, such as a stock or bond, a currency, or even an interest rate or equity index.
- It can also be a real asset or commodity, such as crude oil, gold, or wheat.

Exchange-Traded & OTC Derivatives

- Derivatives trade either on an organized exchange as exchange-traded derivatives or off an exchange as *over-the-counter derivatives* (OTC).
- The OTC network is dominated by banks & financial institutions who trade directly with corporate clients and other financial institutions.
- When a forward-based derivative trades on an exchange, it is typically called a *futures* contract.

Exchange-Traded & OTC Derivatives

- There are several differences between these two:

Exchange Traded	Over-The-Counter

Exchange-Traded & OTC Derivatives

Exchange Traded	Over-The-Counter
Contract Standardization	Terms are agreed to on a contract by contract basis
Clearinghouse acts as third party guarantor ensuring contract's performance to both trading partners	No third party guarantor. Parties must rely on the integrity of each other
Performance bond required	Not necessarily required
Gains and losses accrue on a day to day basis (marking to market)	Gains and losses are settled at the end of the contract
Prices are publicly disseminated immediately	Contracts are private
Heavily regulated	Much less regulated
Delivery rarely takes place	Delivery or final cash settlement usually takes place

Underlying Assets

Commodities:

- wheat, corn, soybeans, sugar, cocoa, coffee,
- crude oil, natural gas, propane
- copper, aluminum, silver, platinum

Financial Assets:

- equity and equity indexes
- interest rates
- currencies

Why Derivatives are Useful

Risk Reduction/Hedging

- Hedging is the attempt to reduce or eliminate the risk of either holding an asset for future sale or anticipating a future purchase of an asset.
- Hedgers start with a pre-existing risk that is generated from a normal course of business.
- Hedging is accomplished by taking a counter or opposite position in the derivative instrument of the asset to be hedged (or one that is very close to it).

Part 2

Option Terminology

Writer:

The seller of the option who is obligated to act.

Strike:

Fixed price at which the rights given to the buyer can be exercised.

Expiry:

The month the contract expires, usually the Saturday following the 3rd Friday after expiration.

Contract:

100 shares of underlying stock.

Option Terminology

Premium:

The price paid for the option.

American-Style Option:

An option that can be exercised anytime up to and including the expiration date.

European-Style Option:

An option that can be exercised only on the expiration date.

Buyer or Holder:

The holder of the option with the right, but not the obligation to act.

Option Terminology

Opening Transaction:

Buying or selling an option to initiate a position.

Long Position:

The buyer or holder of the option.

Short Position:

The seller or writer of the option.

Assigned:

When the long or holder exercises the position.

Option Positions

	HOLDER (Buyer) Pays premium	**WRITER** (Seller) Receives premium
CALL	**right** to **buy** underlying asset at fixed price	**obligation** to **sell** underlying asset at fixed price
PUT	**right** to **sell** underlying asset at fixed price	**obligation** to **buy** underlying asset at fixed price

Option Risks

	HOLDER (Buyer)	**WRITER** (Seller)
CALL	Lose premium	Loss of potential capital gains Unlimited loss if writing a naked call
PUT	Lose premium	Loss if price falls

Options and Leverage

Suppose you believe that the price of a company's stock was going to rise from $50 to $75. You could purchase 100 shares or an option contract.

Buy The Stock

Buys 100 shares at $50	($5,000)
Sell 100 shares at $75	$7,500
Profit	$2,500

Buy One Call Option Contract @ $5

Buy 1 contract (100 x $5)	($ 500)
Sell 1 contract [($75 – $50)] x 100	$2,500
Profit	$2,000

Option Terminology

Intrinsic Value:

The value of certainty. The in-the-money portion of a call or put.

Time Value:

The value of uncertainty. The amount an option is trading above it's intrinsic value.

Option Valuation

When does an option have value?

In-the-money option

Call

Put

Out-of-the-money

Call

Put

Intrinsic Value

Time Value

Option Valuation

In-the-money option:	Call option: Market Price > Strike Price Put option: MP < SP
Out-of-the-money option:	Call option: MP < SP Put option: MP > SP

Intrinsic Value

The in-the-money portion of an option's price.

Time Value

The option premium less intrinsic value.

Option Pricing

Intrinsic Value:

- Call Options = Market Price – Strike Price
- Put Options = Strike Price – Market Price

Time Value:

- Market Value – Intrinsic Value

Option Pricing Examples

The XYZ December 50 call is trading for $7.25 when XYZ's stock price is $56.

- What is the Intrinsic Value?
- What is the Time Value?
- Is the option in-, out-, or at-the-money?

Option Pricing Examples

Intrinsic Value= $56 – $50 = $6

Time Value = $7.25 – $6= $1.25

Since the intrinsic value is positive, the option is in-the-money.

Option Pricing Examples

The ABC June 25 put is trading for $4.85 when ABC's stock price is $21.

- What is the Intrinsic Value?
- What is the Time Value?
- Is the option in-, out-, or at-the-money?

Option Pricing Examples

Intrinsic Value= $25 – $21 = $4

Time Value = $4.85 – $4= $0.85

Since the intrinsic value is positive, the option is in-the-money.

Option Expectations

	HOLDER (Buyer)	**WRITER** (Seller)
CALL	stock price ↑	stock price or no change ↓
PUT	stock price ↓	stock price or no change ↑

Why use a Call?

HOLDER (Buyer)	**WRITER** (Seller)
• Greater leverage vs. buying the stock	• Earn additional income
• Lock in a future price	• Reduce the net cost of the investment
• Protecting a short	

Why use a Put?

HOLDER
(Buyer)

- Less risk than shorting the stock
- Lock in a future price
- Protect an existing position

WRITER
(Seller)

- Earn additional income
- Acquire stock at a lower net price

Call Writing

Covered:

- writer owns the underlying shares
- if assigned must sell the shares to the buyer
- keeps the premium, and may be able to lock in a profit

Naked:

- writer does not own the underlying shares
- highly speculative position
- if assigned, must purchase the securities at a higher price and sell them at the lower strike price

Put Writing

Cash Secured:

- writing a put and setting aside cash equal to the strike price
- invest in a short-term, liquid, money market security: i.e., T-bill
- if assigned buy the shares using the proceeds

Naked:

- no position in the stock and no cash set aside to purchase the shares
- highly speculative position
- wants the price to remain at or above the strike

Options - Summary

You own	Your belief	Your action	Your risk
Call Option	Stock price will increase	Exercise or sell if price rises	Lose cost of investment
Put Option	Stock price will decline	Exercise or sell if price falls	Lose cost of investment
You write	**Your belief**	**Your action**	**Your risk**
Call Option	Stock price will decline	None or can buy back option	Must sell stock at strike price
Put Option	Stock price will increase	None or can buy back option	Must purchase stock at strike price

Options versus Forwards and Futures

Options:

- Are contracts between two parties: a buyer and a seller.
- The buyer has the *right*, but not the obligation, to buy or sell a specified quantity of the underlying asset in the future at a price agreed upon today.
- The seller is *obligated* to complete the transaction if called upon to do so.

Options versus Forwards

Forwards:

- Are contracts between a buyer and a seller.
- Both parties *obligate* themselves to trade the underlying asset in the future at a price agreed upon today.
- Neither party has given the other any right.
- They are both obligated to participate in the future trade.

Primary Differences between Options & Forwards

- An option holder has a *right*, while forwards represent *obligations* to buy or sell in the future.
- Options have a trigger point, known as the exercise price above or below which it has value at expiration.
- Forwards have no such trigger point, they develop value as soon as the market price changes.
- The purchase of an option requires an immediate payment. No money need change hands upon purchase or sale of a forward contract.

Forwards

- A forward contract is one which obligates one party to buy and another to sell a defined amount of an underlying asset at an agreed upon price at a specified time in the future.
- The buyer does not pay the agreed upon price right away, nor does the seller deliver the underlying interest.
- Payment and delivery take place at a specified date in the future known as the delivery date.
- The delivery price is agreed upon when the contract is entered into.
- OTC equivalent of futures contracts that can be customized.

Futures

- A futures contract is an exchange-traded forward agreement between two parties obligating one to buy and one to sell an underlying asset:
  - in a standardized quantity and quality
  - at a price agreed upon today
  - on a future date

Margin Requirements

Initial Margin:

- The original margin required when the contract is entered into.

Maintenance Margin:

- The minimum account balance that must be maintained while the contract is still open.

Both are set by the exchange on which the contract trades. Firms may impose higher rates on their clients.

Margin Requirements

Marking-to-Market:

- The daily settlement of gains and losses.
- Changes in the price of the contract from the previous day will lead either to a payment by the long position to the short, or vice versa.
- Payment is *not* made directly between the long and short, but instead between the counterparties' respective investment dealers through the clearinghouse.

Why Buy Futures Contracts?

- Investors buy futures either to profit from an expected **increase** in the price of the underlying asset, or to lock in a **purchase** price for the asset on some future date.

- **Speculation**:
  Attempting to profit from an expected **increase** in the price of the underlying asset.

- **Risk Management:**
  Locking in a future price for the asset.

Why Sell Futures Contracts?

- Investors sell futures either to profit from an expected **decline** in the price of the underlying asset, or to lock in a **sale** price for the asset on some future date.

- **Speculation**:
  Attempting to profit from an expected **fall** in the price of the underlying asset.

- **Risk Management:**
  Locking in a future price for the asset.

Rights, Warrants, and Options

Rights, warrants, and call options share certain characteristics. Each gives the holder or buyer the right to purchase additional stock at a specified price until an expiry date.

How do rights and warrants differ from call options?

Rights

- A **right** is a privilege granted to an existing shareholder to acquire additional shares directly from the issuing company.
- To raise capital by issuing additional common shares, a company may offer shareholders the *right to buy* shares in direct proportion to the number of shares they already own.
- For example, the offer may be based on the right to buy one additional share for each ten shares held.

Intrinsic Value of a Right

Cum rights period

$IV_{cum} = (S - X)/(n + 1)$

Ex-rights period

$IV_{ex} = (S - X)/n$

Where:

- IV_{cum} = Value of the right in the cum-rights period
- IV_{ex} = Value of the right in the ex-rights period
- S = Market price of the stock
- X = Exercise or Subscription price of the right
- n = Number of rights required to purchase 1 new share

Rights Exercise

ABC Inc. declares a rights offering. Shareholders of record on Friday, July 16 were granted one right for each common share held. Five rights needed to subscribe for each new ABC common share at a subscription price of $23. The rights expired at the close of business on August 18. Share prices were as follows:

Monday	July 12	$26.50
Tuesday	July 13	$26.00
Wednesday	July 14	$25.50
Thursday	July 15	$26.15
Friday	July 16	$26.10
Monday	July 19	$26.35
Tuesday	July 20	$26.75

Rights Exercise

1. When will the shares begin to trade ex-rights?
2. What is the intrinsic value of a right on the last day of the cum-rights period?
3. What is the intrinsic value of a right on the first day of the ex-rights period?

Rights Exercise

When will the shares begin to trade ex-rights?

Two business days before the record date.

Since the record date was Friday, July 16, the shares would begin to trade ex-rights on Thursday, July 15.

Rights Exercise

What is the intrinsic value of a right on the last day of the cum-rights period?

The last date of the cum-rights period is Wednesday, July 14.

Value:

$$\frac{S - X}{n + 1} = \frac{\$25.50 - 23}{5 + 1} = \$0.4266$$

Rights Exercise

What is the intrinsic value of a right on the first day of the ex-rights period?

The first day the shares trade ex-rights is Thursday, July 15.

Value :

$$\frac{S - X}{n} = \frac{\$26.15 - 23}{5} = \$.63$$

Warrants

- A warrant provides the holder with an option to buy shares in a company from the issuer at a set price for a set period of time.
- Warrants are often attached to new debt and preferred issues to make these issues more attractive to buyers by giving the buyer of a new issue the opportunity to participate in capital gains on the common share's market price.

Warrant Math

ABC Co. warrants entitle the owner to buy one share at an exercise price $40. The warrant has a market value of $5 and the common are trading at $42 a share.

What is the intrinsic value of the warrant?

What is the time value of the warrant?

Warrant Math

Intrinsic Value of a Warrant

= (Market Price Per Share – Exercise Price)

= ($42 – $40)

= $2

Time Value of a Warrant

= Market Price of the Warrant – Intrinsic Value

= $5 – $2

= $3

Warrant Math

Assume in this case that the common trade at $35 a share (all other factors remain the same).

Calculate the intrinsic value

IV = $35 – $40 = 0

Calculate the time value

TV = $5 – 0 = $5